[METRIC]

A-A-59315/22
September 4, 1998
SUPERSEDING
MIL-P-85856/22
December 7, 1988

COMMERCIAL ITEM DESCRIPTION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS SILICON-ALUMINUM PLUS POLYMER, TYPE I, CLASS 1

The General Services Administration has authorized the use of this commercial item description sheet as a replacement for MIL-P-85856/22 by all federal agencies.

The complete requirements for procuring the powder material described herein shall consist of this document and the issue in effect of commercial item description A-A-59315/GEN.

1. SCOPE. This commercial item description sheet (CID) covers a silicon-aluminum plus polymer powder that is applied using either plasma arc or combustion gas (flame spray) techniques.

2. SALIENT CHARACTERISTICS.

2.1 Chemical composition. Chemical composition shall conform to the following percentages, by weight, as shown in table I.

2.2 Particle size distribution. The particle size distribution shall conform to table II.

3. REGULATORY REQUIREMENTS. Regulatory requirements shall be in accordance with A-A-59315/GEN.

4. QUALITY ASSURANCE PROVISIONS. Quality assurance provisions shall be in accordance with A-A-59315/GEN.

5. PACKAGING. Packaging shall be in accordance with A-A-59315/GEN.

6. NOTES. The notes specified in A-A-59315/GEN are applicable to this CID sheet.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3, Highway 547, Lakehurst, NJ 08733-5100.

FSC 3439

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TABLE I. Chemical composition. 1/

Element	Percentage	
	min.	max.
Iron	--	0.8
Aluminum	Balance	
Silicon	11.0	13.0
Copper	--	0.3
Others		
Each	--	0.2
Total	--	0.6

1/ Percentages by weight are for the metallic component only. Material shall be a blend of 59-61 percent by weight of a silicon-aluminum alloy and 39-41 percent by weight of a completely aromatic polymer system. The polymer shall conform to the following particle size distribution before blending: A minimum of 99 percent by weight shall pass through a No. 100 (150μm) sieve; a minimum of 85 percent by weight shall pass through a No. 170 (90μm) sieve; and a maximum of 25 percent by weight shall pass through a No. 325 (45μm) sieve.

TABLE II. Particle size distribution by weight (percent).

Sieve Designation		Weight passed through sieve	Weight retained on sieve
Mesh	Microns (μm)		
200	75	85.0 min.	--
325	45	70.0 to 90.0	--
	10	10.0 max.	--

CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
Army - MR
Navy - AS
Air Force - 11
DLA - GS

Review activities:
Navy - SH
Air Force - 71, 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA/FSS-7FXE

Preparing activity:
Navy - AS
(Project 3439-0932)

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